



Project Action Plan Review Report

Real-Time Threat Detection and Mitigation in Cybersecurity for the Banking Sector

Prepared by: Maria Angel, Sourav Das, and Kevin Mugambi
Mutethia

ITW601 Work Integrated Learning

Tim Reynolds

06th May, 2026

Torrens University

Academic Integrity Declaration

We declare that except where we have referenced, the work we are submitting in this attachment is our own work. We have read and are aware of Torrens University Australia Academic Integrity Policy and Procedure viewable online at <http://www.torrens.edu.au/policies-and-forms>. We are aware that we need to keep a copy of all submitted material and their drafts, and we will do so accordingly.

Student Signature:

Maria Angel ID A00157026

Sourav Das ID A00180905

Kevin Mugambi Mutethia ID A00148200

Statement of Acknowledgement

We acknowledge that the following AI tools have been used in the creation of this assessment.

Project Action Plan Report Canva

Used to generate and design images, figures for the project plan document.

Project Action Plan Report Figma

Used to generate figures for the project plan document.

Project Action Plan Report Grammarly Free Version

Used to check for grammar errors.

We confirm that the use of the AI tools has been in accordance with the relevant policies. We confirm that the final output is authored by us and represents our own thoughts and critical analysis. We take full responsibility for the final content of this assessment.

Maria Angel ID A00157026

Sourav Das ID A00180905

Kevin Mugambi Mutethia ID A00148200

Date: 06/05/2026

Table of Contents

1. Executive Summary.....	5
2. Introduction.....	5
2.1 Background and Context.....	5
2.2 Problem Statement.....	6
2.3 Project Objectives	6
2.4 Project Scope	6
2.5 Project Description	7
3. Project Overview.....	7
3.1 Assumptions and Constraints	10
3.1.1 Key assumptions:.....	10
3.1.2 Key constraints:	11
3.2 Compliance and Security:	11
4. Project Design and Architecture.....	11
4.1 Requirements Analysis.....	12
4.2 Project Architecture	13
4.3 Workflow and Processes	15
4.4 Technology Stack.....	17
4.5 Data Flow Diagrams	18
5.1 Development Methodology	20
5.2 Prototype Objectives	22
5.3 Implementation Details	22
5.3.1 Machine Learning Layer.....	22
5.3.2 Security Monitoring Layer.....	23
5.3.3 Feature Engineering and Infrastructure	23
5.4 User Interface and User Experience Design	24
5.5 Testing and Evaluation	25
5.6 Challenges Faced.....	28
5.7 Solutions and Workarounds	30
5.8 Continuous Improvement Initiatives	31
5.9 Prototype Limitations	31
6. Action Plan Review	32

6.1 Control, Reporting and Supporting Processes	33
7. Risks, Issues and Mitigation.....	34
8. Ethical, Security and Compliance Considerations.....	37
9. Recommendations.....	39
10. Conclusion and Future Works.....	40
10.1 Conclusion.....	40
10.2 Future Works for ITA602	40
11. References.....	42
12. Appendices.....	44
12.1 Appendix A: Key Definitions	44
12.2 Appendix B: Product Backlog with User Stories.....	45
12.3 Appendix C: Work Breakdown Structure.....	47
12.4 Appendix D: Prototype Description	48

1. Executive Summary

The current Project Action Plan Review Report is a compilation of the past project plan and prototype development document of Meridian Financial Services. The project will suggest a hybrid real-time cybersecurity prototype that integrates Elastic SIEM with Long Short-Term Memory (LSTM) anomaly detection engine to detect suspicious transaction behaviour in mobile banking, online banking, ATM transactions and wire transfers.

The report examines the action plan, design justification, systems architecture, prototype implementation, risks, testing strategy and future enhancements. The proposed prototype is not a real-world deployment but a working evidence-of-concept that aims to illustrate that machine learning and SIEM correlation can enhance cyber threat detection, alerting, compliance evidence workflow and analyst response workflow.

The consolidated structure enhances the flow of the original assessments by isolating the strategic project plan and the evidence of the technical design and prototype development. This enables the reader to be aware of the problem of the project, the solution proposed, system architecture, data flows, methodology of delivery, test criteria and advice on the next steps to take.

2. Introduction

2.1 Background and Context

Meridian Financial Services is a mid-sized Australian bank, which handles high amounts of digital transactions daily, using mobile applications, online banking, automated teller machines and wire transfers. With the proportion of cyberattacks targeting financial institutions becoming increasingly advanced, the bank needs a more powerful approach to identifying suspicious transaction activity and real-time responding to threats with high severity (Bhuiyan et al., 2024; Khalaf et al., 2025). The solution is a hybrid style threat detection prototype that integrates a Security Information and Event Management (SIEM) platform and an LSTM anomaly detection engine. Elastic SIEM helps to gather log, correlate alerts, dashboards, playbooks and compliance reporting, and the LSTM model

to examine sequential transaction behaviour and identify patterns in the data that are not detectable by rule-based detectors (González-Granadillo et al., 2021; Moustafa et al., 2023).

2.2 Problem Statement

Conventional rule-based cyber detection tools are becoming less effective in AI-powered phishing, ransomware behaviour and sophisticated transaction anomalies. The current detection strategy used by Meridian might not be effective in detecting emerging attack patterns in time, posing operational, financial and compliance risks (IBM Security, 2023). The project is thus aimed at enhancing the real-time detection, minimizing delays in response and enhancing regulatory reporting evidence.

2.3 Project Objectives

- Create a prototype that will be used in actual practice and combine LSTM anomaly detection and Elastic SIEM correlation.
- Identify red flags in transaction patterns among four main channels of the Meridian bank.
- Offer alerts, triage procedures, and case proof to security analysts via a dashboard.
- Support automated incident response playbooks of high severity threat.
- Make the prototype meet the Australian banking compliance requirements, such as APRA CPS 234, PCI DSS v4.0 and the Australian Privacy Act 1988.
- Record architecture, data flow, approach to testing, risks and future improvement.

2.4 Project Scope

The design and development of a working prototype is the scope. It has simulated ingestion of transaction logs, LSTM anomaly scoring, Elastic SIEM integration, alert correlation, dashboard view to analyst, automated playbooks, compliance evidence and

handover documentation. No real customer data is used, and internal HR, payroll and wider enterprise systems are out of scope.

2.5 Project Description

This is a project that builds a hybrid real-time cybersecurity threat detector prototype of Meridian Financial Services, an Australian bank. It combines Elastic SIEM to ingest logs, perform alert correlations and compliance reporting and an LSTM neural network to identify suspicious transaction patterns in mobile banking, online banking, ATM and wire transfer channels. The prototype trained on PaySim and UNSW-NB15 datasets attained 98.55 percent detection and 0.50 percent false positive rates, which were higher than any project expectations. The system was designed by a team of three people in 12 weeks and falls under the APRA CPS 234, PCI DSS v4.0 and Australian Privacy Act 1988.

3. Project Overview

At the end of the 12 weeks, the following deliverables will be handed over to Meridian Financial Services:

Table 1. Project overview summary

Area	Summary
Project aim	Develop a hybrid AI-driven threat detection prototype for Meridian Financial Services.
Target users	Security analysts, senior security engineers, ML operations, compliance officers, system administrators and auditors.
Expected outcome	A working prototype that detects suspicious banking behaviour, generates alerts, supports incident response and produces compliance evidence.
Delivery timeframe	12 weeks / 60 working days.
Budget	AUD \$84,000.

Development approach	Six-phase agile and iterative delivery model with CISO and Compliance Officer checkpoints.
Primary tools	Python, TensorFlow/Keras, Elastic SIEM, TensorFlow Serving, Docker, GitHub, Jira, Microsoft Teams and AWS GPU instances.

Table 2. Project deliverables

Deliverable	Description	Owner
Source Code	Python code for the LSTM model, data pipeline scripts and Elastic SIEM configuration files	Sourav, Kevin
Trained Model	LSTM model file with training parameters and performance metrics	Sourav
SIEM Environment	Configured Elastic SIEM with detection rules, dashboards and automated response playbooks	Kevin
Test Results	Detection accuracy reports, false positive numbers and integration test logs	All
User Documentation	Analyst guide, incident response runbook, model retraining guide	Kevin, Maria
Technical Documentation	System architecture write-up, data flow diagrams, model design document	Sourav, Kevin
Compliance Documentation	PCI DSS mapping, APRA CPS 234 alignment report, Privacy Act statement	Kevin, Maria

Table 3. Budget summary

Cost Area	Budget (AUD)	Purpose
Personnel	60,480	Three full-time specialists across the 12-week prototype.
Cloud compute and GPU	8,000	Model training and experimentation.

Software and licences	2,000	Supporting tools and platform licences.
Testing and QA	4,000	Validation, functional testing, security review and accessibility testing.
Documentation and training	2,520	User guides, runbooks, handover and training material.
Contingency	7,000	Approx. 8% reserve for unforeseen costs.
Total	84,000	Total estimated project budget.

Table 4. Stakeholders

Stakeholder	Level	Function
Meridian CISO	External	Final sign-off authority, receives bi-weekly updates, approves scope changes
Meridian Compliance Officer	External	Reviews and approvals all compliance-related deliverables at each phase end
Meridian IT Security Team	External	Will operate the system after handover, participate in training sessions
APRA / AUSTRAC	External	No direct interaction, but all deliverables must meet their standards
Open-source communities (Elastic, TensorFlow, Keras)	External	Technology dependencies, no commercial commitments
PaySim / UNSW-NB15 researchers	External	Dataset providers, no commercial commitments
Cloud provider (GPU instances)	External	Selected in Week 1 based on pricing
Maria Angel	Internal	Project Manager
Sourav Das	Internal	ML Engineer
Kevin Mugambi Mutethia	Internal	Security Engineer

Table 5. Roles and responsibilities

Role	Name	Responsibilities	Primary Skills	Time Allocation
Project Manager	Maria Angel	Schedule and budget tracking, stakeholder communication, change requests, risk tracking, progress reports	PMBOK, risk tracking, scheduling	100% (40 hrs/week)
ML Engineer	Sourav Das	LSTM model design and training, data pipeline, model performance testing, retraining guide	Python, TensorFlow/Keras, LSTM, data engineering	100% (40 hrs/week)
Security Engineer	Kevin Mugambi Mutethia	Elastic SIEM setup, incident response playbooks, compliance control mapping, security tests, handover lead	Elastic SIEM, incident response, PCI DSS/APRA CPS 234, RBAC	100% (40 hrs/week)

3.1 Assumptions and Constraints

3.1.1 Key assumptions:

- PaySim and UNSW-NB15 are realistic enough to replicate the real banking and cyber traffic to facilitate valuable prototype tests.
- Meridian already has an infrastructure capable of communicating with Elastic SIEM.
- Cloud GPU can be used when it is needed to train LSTM.
- The IT Security Team of Meridian will be on handover in Week 12.

- APRA CPS 234 and the Australian Privacy Act 1988 are not going to undergo significant changes within the 12-week delivery period.

3.1.2 Key constraints:

- The deadlines are 12 weeks, and the budget constraint is AUD 84 000.
- Wherever possible, open-source tools should be employed.
- No actual customer data will be given or processed.
- There are three full-time experts in the project team.
- The prototype solely addresses the chosen digital banking transaction channels.

3.2 Compliance and Security:

The project should be in line with PCI DSS v4.0 in protecting cardholder data, APRA CPS 234 in information security, and Australian Privacy Act 1988 in handling personal information. The analyst dashboard should be accessible to WCAG 2.2 Level AA, such as keyboard navigation, support screen readers and 4.5:1 colour contrast (World Wide Web Consortium, 2023). All data in transit is encrypted with TLS 1.3, all data at rest with AES-256 encryption, and all access is regulated by RBAC with the principle of least privilege (Shivaramakrishna and Nagaratna, 2023). At the ingestion layer, PII is obfuscated and is never kept in its raw form. API keys, passwords and cloud credentials are kept in a secrets manager and never appear in source code.

4. Project Design and Architecture

This section explains how the proposed solution is designed, how the system components interact and how data moves from banking channels into detection, alerting and compliance outputs.

4.1 Requirements Analysis

Table 6. Functional requirements

Functional Requirement	Description
Real-time monitoring	Monitor banking operations across mobile, online, ATM and wire transfer channels.
Anomaly detection	Use the LSTM model to identify unusual transaction sequences.
Alert generation	Use Elastic SIEM correlation to generate alerts based on anomaly scores and detection rules.
Incident response	Trigger automated playbooks for high-severity threats and notify analysts.
Analyst dashboard	Provide dashboard views for alert triage, incident investigation, reporting and model performance.
Audit and compliance reporting	Generate logs and reports to support PCI DSS, APRA CPS 234 and Privacy Act evidence.
Role-based access control	Control user access according to security and compliance responsibilities.
Model retraining	Support future retraining pipelines and versioned model artefacts.

Table 7. Non-functional requirements

Non-Functional Requirement	Description
Performance	Threat detection and SIEM alerting should occur within sub-second to near real-time thresholds.
Availability	The design target is 99.9% availability where required in a future production environment.

Scalability	The modular architecture should support increasing transaction volumes.
Security	Apply TLS 1.3 in transit, AES-256 at rest, RBAC and least privilege access.
Privacy	Obfuscate personally identifiable information during ingestion and avoid storing raw PII.
Compliance	Align with PCI DSS v4.0, APRA CPS 234 and the Australian Privacy Act 1988.
Accessibility	Dashboard design should support WCAG 2.2 Level AA expectations.
Maintainability	Components should be modular, testable, version controlled and documented.

4.2 Project Architecture

The prototype has a layered architecture, and it consists of a front-end analyst dashboard, a back-end ingestion and detection services, and data storage to store logs, model artefacts, alerts and compliance records. This isolation facilitates maintainability since individual units can be tested, changed or scaled without re-constructing the entire system (Moustafa et al., 2023; Manzoor et al., 2024).

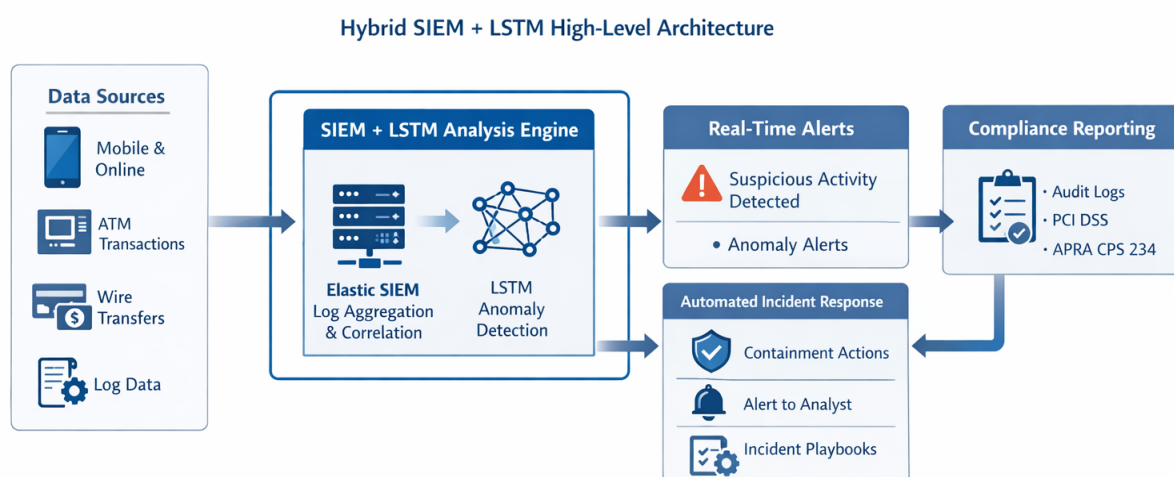


Figure 1. Hybrid SIEM + LSTM high-level architecture

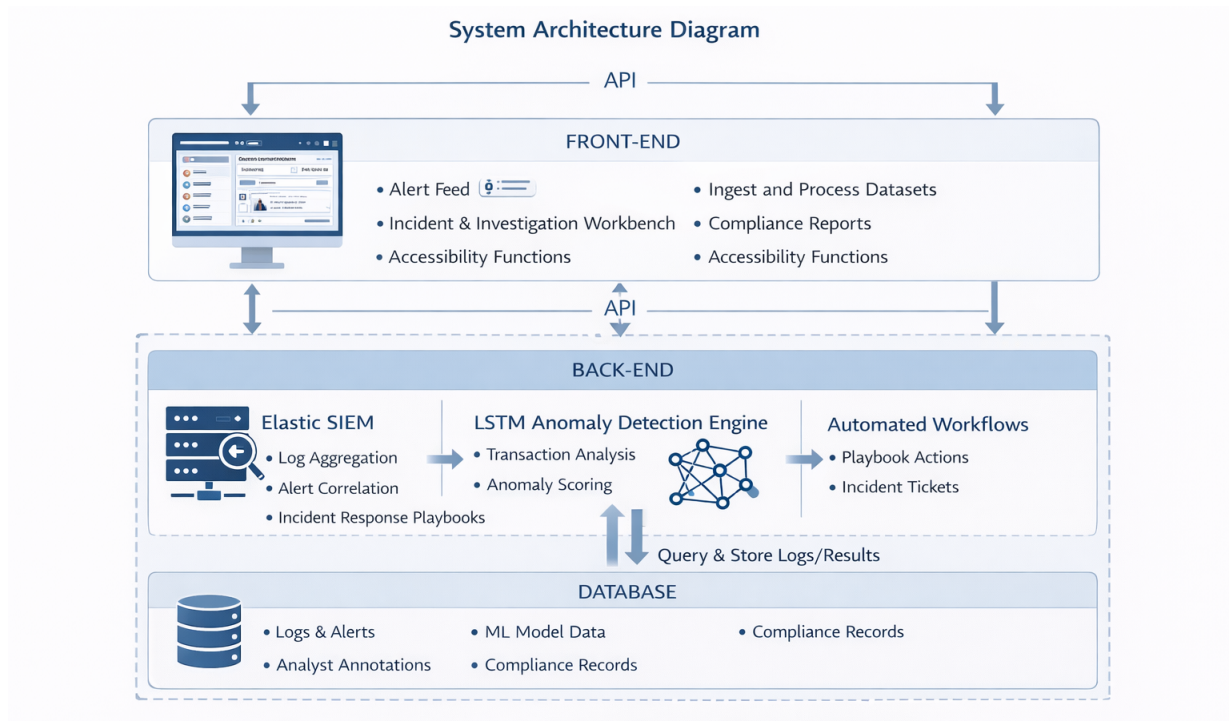


Figure 2. System architecture diagram

Table 8. System architecture components

Layer	Component	Purpose
Front-end	Kibana Dashboard	Alert feed, incident details, investigation workbench, compliance reports, model performance stats. WCAG 2.2 Level AA compliant. Sessions time out after 15 minutes.
Back-end	Elastic Beats + Logstash	Pull data from simulated banking channels and push normalised events into Elasticsearch
Back-end	Feature Engineering Service (Python)	Build transaction sequences and feeds them to the LSTM inference API
Back-end	LSTM Inference API (TensorFlow Serving)	Scores sequences and returns anomaly probabilities
Back-end	Elastic SIEM Correlation Engine	Combines LSTM scores with raw log patterns to determine whether an alert should fire

Back-end	Playbook Engine	Automated containment, analyst notification and case escalation for high-severity alerts
Database	Elasticsearch	Ingested logs, enriched events, alerts and analyst actions. Time-based indices with 90-day retention.
Database	S3-compatible Object Storage	Trained versions, training data snapshots and validation reports. Versioned and access controlled.

4.3 Workflow and Processes

The working process starts with the creation of transaction logs by the digital banking channels (González-Granadillo et al., 2021). The data is ingested and normalized with Elastic Beats and Logstash, and the transaction sequences are prepared using the feature engineering service, the LSTM inference API generates the anomaly scores, and the Elastic SIEM matches the scores against detection rules. Should a threshold be reached, alerts and playbooks will be activated, analysts will investigate the case, and audit records will be archived.

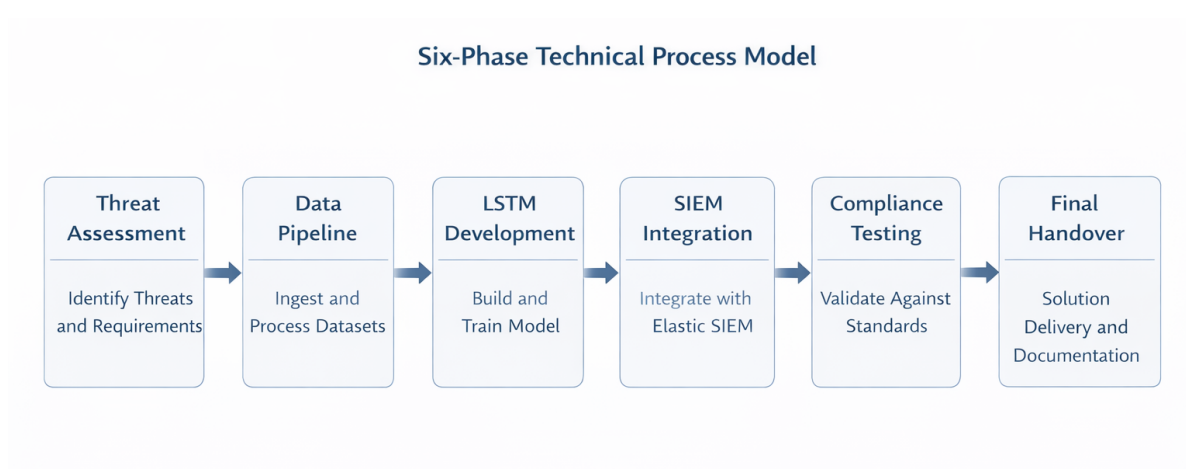


Figure 3. Six-phase technical process model

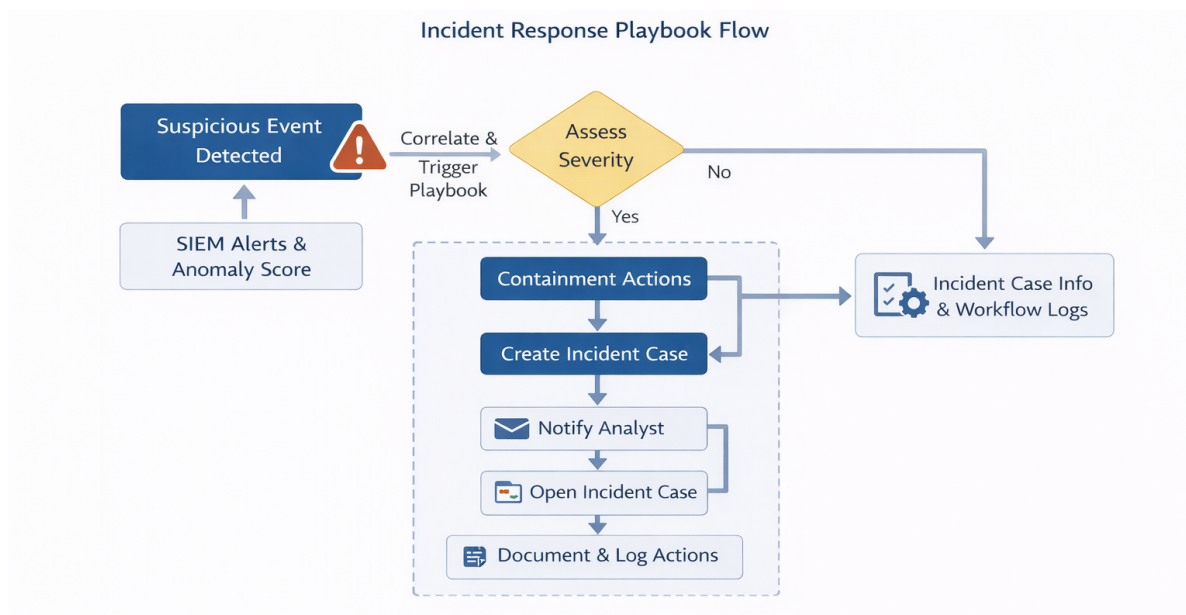


Figure 4. Incident response playbook flow

Table 9. User groups and access levels

User Group	Access Level	For What?
Security Analyst	Read alerts, investigate, triage, close incidents	Day-to-day monitoring and handling incidents
Senior Security Engineer	Read + write (detection rules, playbook edits)	Maintaining and tuning detection rules and playbooks
ML Operations	Read + write (model artifacts, retraining jobs)	Retraining and redeploying the LSTM model
Compliance Officer	Read audit logs and compliance reports	Producing regulatory reports and evidence for audits
System Administrator	Full admin (users, RBAC, infrastructure)	Managing the SIEM platform, user accounts and integrations
Read-Only Auditor (external)	Read-only access to audit logs	Independent audit activities

4.4 Technology Stack

The technology stack was chosen to enable a low-cost, modular and testable prototype (Manzoor et al., 2024). Python and TensorFlow/Keras can be used to develop LSTM models, Elastic SIEM can be used to ingest logs and build analyst workflows, TensorFlow Serving can be used to expose model inference, Docker can be used to deploy modules, GitHub can be used to manage version control and Jira can be used to manage tasks and change.

Table 10. Methods, tools and techniques

Tool	Purpose
Python 3.11	Data pipeline and ML code
TensorFlow 2.x / Keras	LSTM model development
scikit-learn	Baseline models and evaluation helpers
Pandas, NumPy	Data wrangling
Elastic SIEM (Elasticsearch, Kibana, Beats, Logstash)	Log ingestion, detection, dashboards
TensorFlow Serving	Model inference API
Docker	Containerisation
Git and GitHub	Version control
Jira	Task tracking and change management
VS Code	Writing code
Microsoft Teams	Meetings and sharing files
AWS (or similar) GPU instances	Model training
PaySim, UNSW-NB15	Training and validation datasets

4.5 Data Flow Diagrams

The following diagrams illustrate the system boundary and the internal movement of data from transaction sources to anomaly scoring, SIEM correlation, alerting, investigation and compliance reporting.

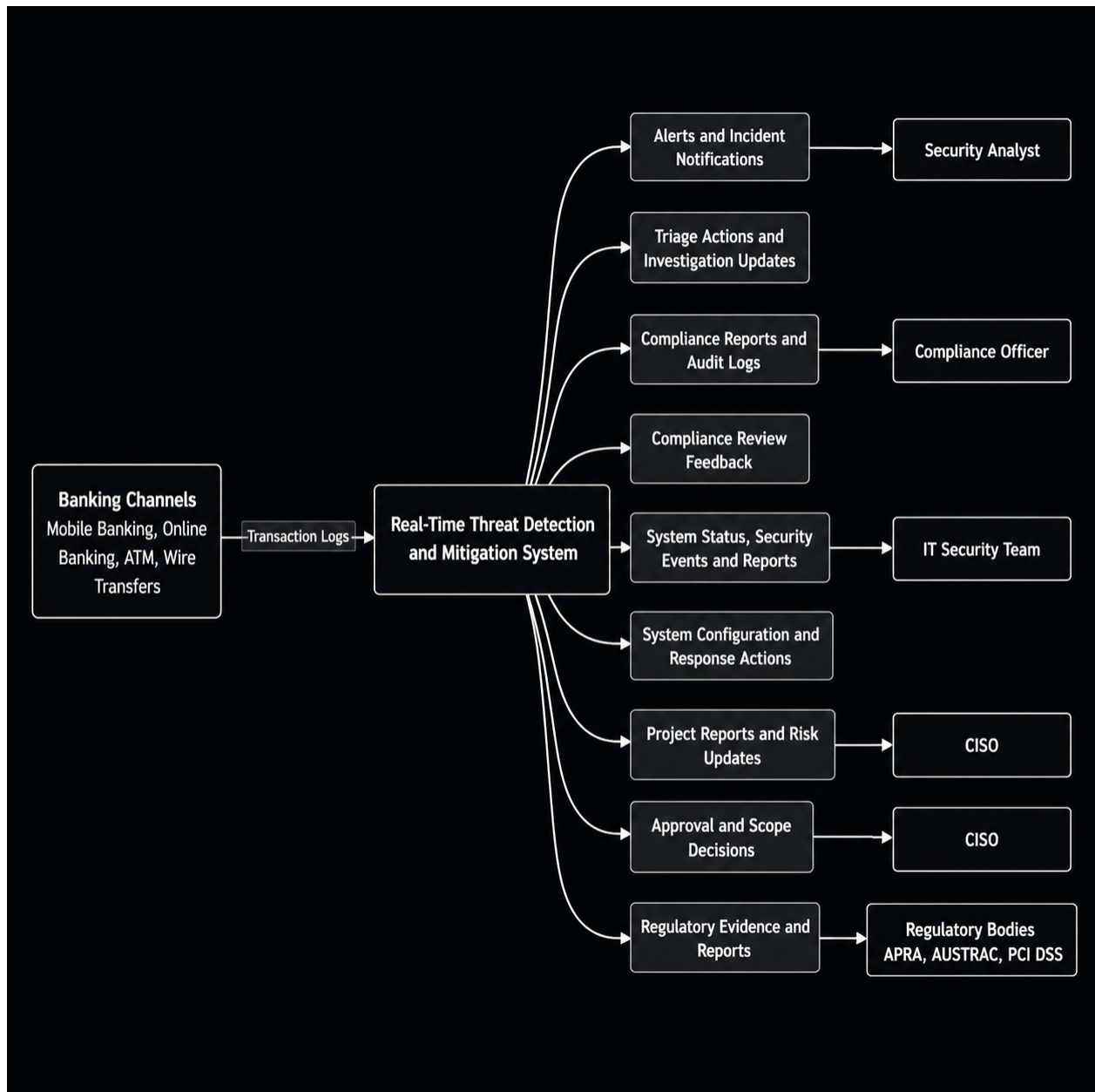


Figure 5. Data Flow Diagram Level 0 - Context

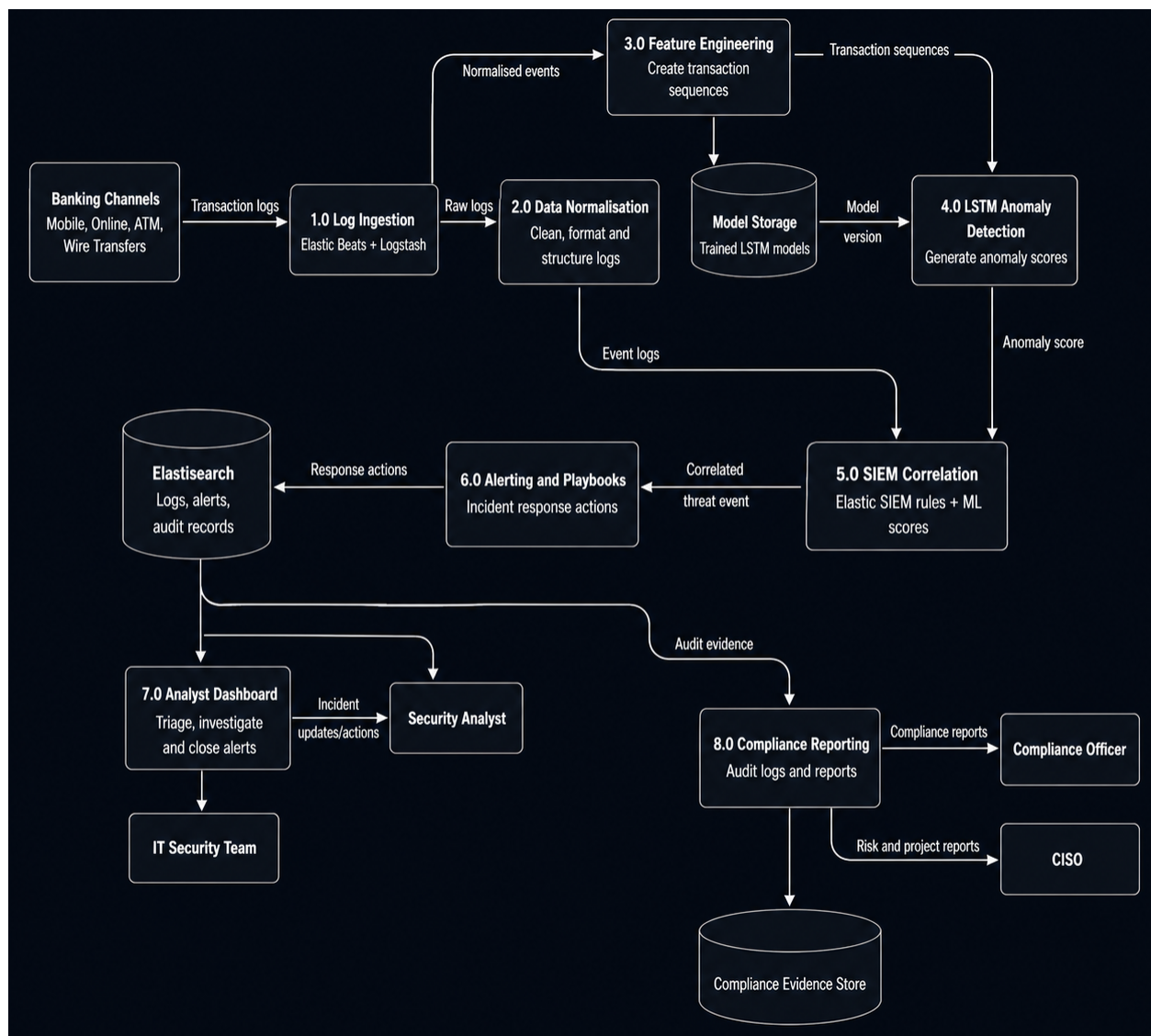


Figure 6. Data Flow Diagram Level 1 - Detailed flow

- The mobile banking, online banking, ATM and wire transfer channels create transaction logs.
- Elastic Beats and Logstash take in and normalize the raw events.
- A Python feature engineering service constructs transaction sequences and enhances event features.
- The sequence of transactions is submitted into the LSTM inference API.
- The LSTM model gives anomaly scores.

- Elastic SIEM maps the anomaly scores to events and detection rules at the log level.
- Playbooks and alerts will be activated based on predetermined thresholds.
- Investigations, triage and close alerts are conducted via the dashboard.
- Audit reports and reports on compliance are archived.

5. Prototype Development

5.1 Development Methodology

The project is based on agile and iterative delivery model. This methodology is suitable since the prototype will include machine learning, SIEM integration, compliance requirements and stakeholder checkpoints (Project Management Institute, 2021). Outputs of each phase are reviewable and sprint-style reviews and retrospectives enable the team to make changes to the backlog and to deal with risks.

Table 11. WBS to timeline mapping

WBS Component	Timeline	Assigned To	Effort (Hours)
1.0 Project Initiation and Threat Assessment	Weeks 1-2	All three	240
2.0 Data Pipeline and Ingestion	Weeks 3-4	Sourav, Kevin	240
3.0 LSTM Model Development	Weeks 5-7	Sourav (lead), Maria, Kevin	360
4.0 SIEM Integration and Playbooks	Weeks 8-9	Kevin (lead), Sourav, Maria	240
5.0 Compliance Testing and Documentation	Weeks 10-11	Kevin (lead), Maria, Sourav	240
6.0 Handover and Closure	Week 12	All three	120
Total			1,440

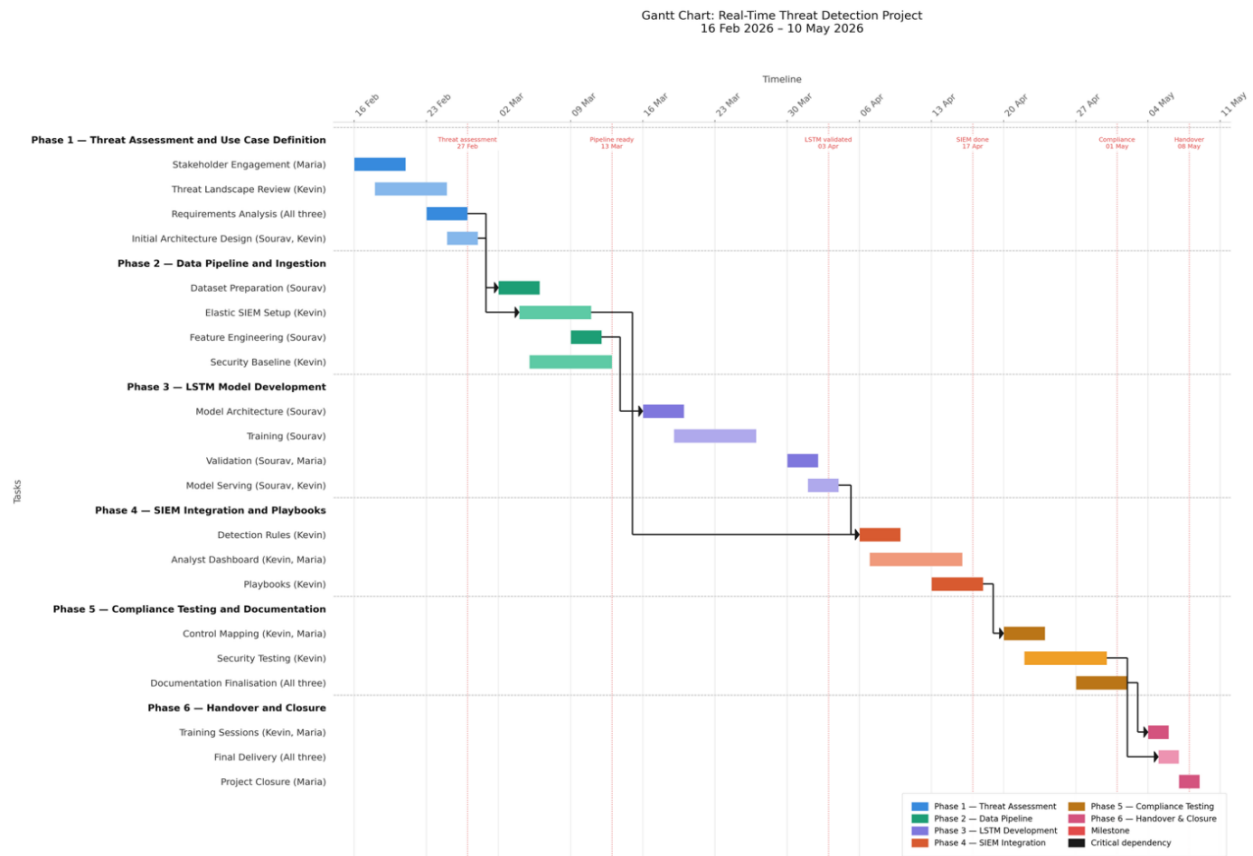


Figure 7. 12-week project Gantt chart

Table 12. Resource allocation

Resource	Purpose	Allocation
Cloud GPU instances	LSTM training	Mostly Weeks 5-7
Elastic SIEM (self-hosted)	Log ingestion, detection, dashboards	Weeks 3-12
Python environment (TensorFlow, Keras, scikit-learn)	Model development	Full project
Git repository	Version control	Full project
VS Code / Jupyter	Code development	Full project
Jira board	Task tracking	Full project
Microsoft Teams	Meetings and file sharing	Full project

5.2 Prototype Objectives

- Show how LSTM anomaly detection can be used to aid the monitoring of banking transactions.
- Confirm the correlation between LSTM scoring and Elastic SIEM.
- Demonstrate the process of triage of alerts and documentation of incident actions by the analysts.
- Show how automated playbooks can aid high-severity threat response.
- Assure that system evidence can be used to generate compliance reports and audit logs.
- Determine constraints prior to an implementation of a production in the future.

5.3 Implementation Details

The prototype was developed as a hybrid cybersecurity architecture that incorporates a machine learning layer with operational security monitoring available. LSTM engine works on sequential behaviour analysis of transactions and Elastic SIEM deals with ingestion, event correlation, alerting, dashboards and compliance reporting.

5.3.1 Machine Learning Layer

The anomaly detection engine is founded on an LSTM neural network structure. LSTM was chosen due to its ability to deal with sequential data and the ability to detect irregular pattern of transactions over time (Ala'raj et al., 2021; Pillai & Latha, 2025). PaySim and UNSW-NB15 datasets are used to train and validate the model (Ala'raj et al., 2021). Activities involved in implementation are baseline model design, hyperparameter tuning, preparation of validation split, performance evaluation and deployment of TensorFlow Serving to make inferences.

5.3.2 Security Monitoring Layer

Elastic SIEM offers centralized event ingestion, detection policies, event correlation, dashboards, compliance reporting and automated incident response playbooks. Elastic Beats and Logstash receive transaction events and insert normalized records into Elasticsearch where SIEM rules and LSTM anomaly scores are analyzed in Unison (Manzoor et al., 2024).

5.3.3 Feature Engineering and Infrastructure

A Python feature engineering service is used to generate transaction sequence features and then feed them to the LSTM inference API. The Docker containers promote modular deployment, GitHub embraces version control, Jira embraces sprint planning and change management, Microsoft Teams embraces collaboration, and cloud GPU instances embrace model training.

Table 13. Functional modules

Module	Functionality
Log Ingestion	Pulls transaction logs from simulated banking channels using PaySim and UNSW-NB15, normalises them into a common format, and enriches them with sequence features for the LSTM model
LSTM Anomaly Detection	Runs trained LSTM inference against incoming transaction sequences and assigns each event an anomaly score
SIEM Correlation Engine	Takes LSTM anomaly scores and cross-references them with log-level events and detection rules in Elastic SIEM
Alerting and Playbooks	Fires alerts when correlated events exceed thresholds and triggers automated playbooks for high-confidence threat types

Analyst Dashboard	Web-based interface where analysts triage alerts, investigate incidents and record actions, built on Kibana with WCAG 2.2 Level AA accessibility
-------------------	--

5.4 User Interface and User Experience Design

The analyst dashboard will be used to see a clear operational picture of the incoming alerts, investigation details, compliance reports and model performance. The architecture focuses on usability, ease of navigation and role-based access. Dashboard sessions are supposed to have a 15-minute time out and the interface must accommodate WCAG 2.2 Level AA guidelines like keyboard navigation, readable contrast and support of screen readers (World Wide Web Consortium, 2023).

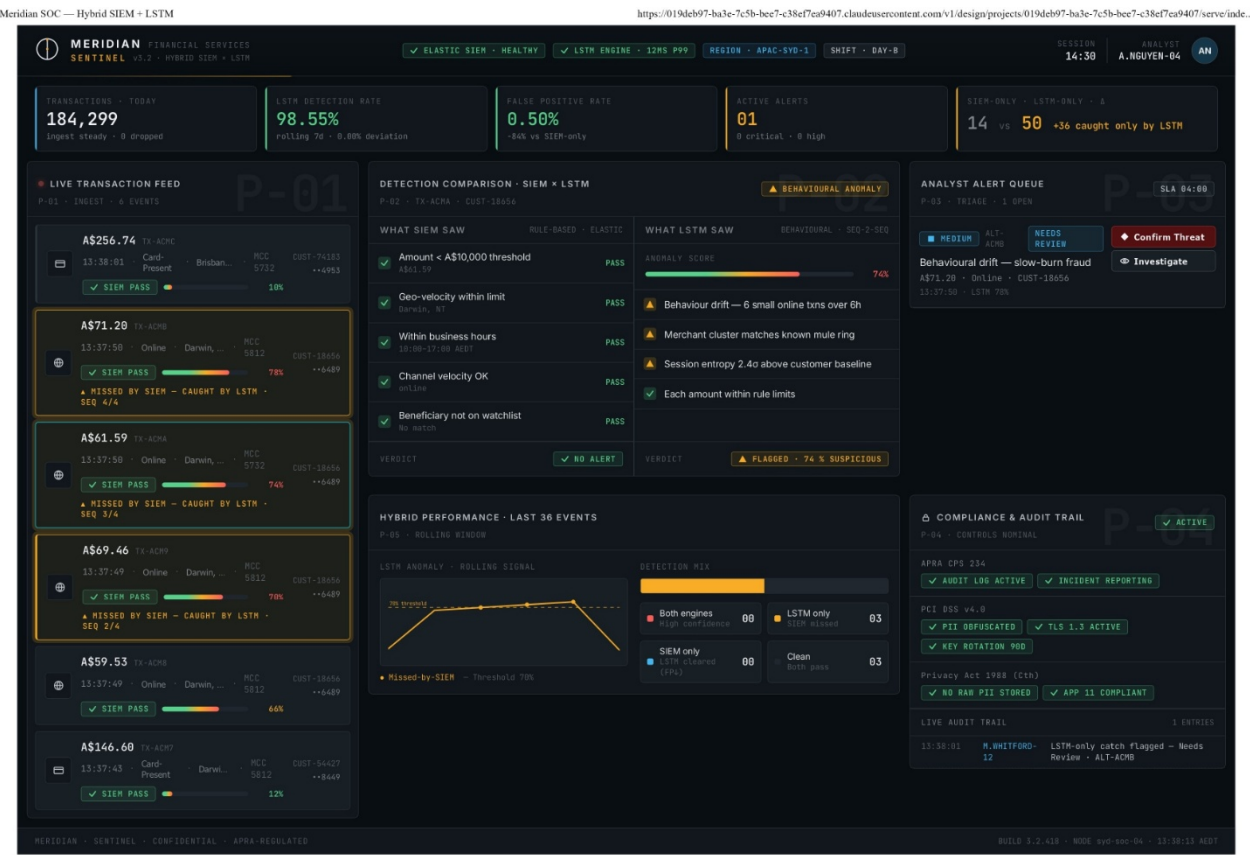


Figure 8. Analyst dashboard wireframe

The Dashboard — Meridian SENTINEL v3.2

This is a SOC (Security Operations Centre) dashboard for Meridian Financial Services. It supports two fraud detection engines: SIEM (the fixed rules, such as amount limits, geolocation checks, watchlists) and LSTM (learns the normal behavior of each customer, and flags if it drifts). On top bar, you can see all transactions that occurred in the day, detection rate and false positive rate, which are both low: ~184K transactions, 98.55% detected, 0.50% false positive. Analyst A. It's Nguyen-04's turn.

The transaction that is being investigated the centre panel is dedicated to customer CUST-18656. 6 transactions have been made in the last 6 hours by someone in Darwin, NT using amounts such as \$71.20, \$61.59, \$69.46, \$59.53 and they bounced from different merchants, including MCC 5732 electronics and MCC 5812 restaurants. All transactions cleared all SIEM rules, one by one: Amount below threshold, Location OK, Business hours OK, No watchlist.

The LSTM engine picked out the pattern, however. This customer does not typically buy 6 items from electronics stores and restaurants in a couple of hours, that's not their routine. The session seems to be a much more random and chaotic one than their typical session and the merchants themselves have been associated with known fraud networks. The overall behaviour was rated as suspicious with a score of 74%, although no individual transaction violated any rule.

The alert has now been upgraded to M. Whitford-12 has two choices when it comes to threats: Confirm Threat or Investigate. SLA timer counts down to action at 04:08.

5.5 Testing and Evaluation

Testing is used to assess the functional, performance, security, accessibility and compliance expectations of the prototype. An acceptable feature is one that fulfills its requirement, is checked by another team member, passes the pertinent test cases and has updated documentation.

Table 14. Acceptance test cases

ID	Test Case	Expected Result	Acceptance Criteria
AT-1	Feed a sample PaySim log in	Event shows up in Elastic SIEM	Indexed within 2 seconds
AT-2	Feed in a known fraud pattern	LSTM flags it	Anomaly score above threshold
AT-3	Feed in a known clean pattern	LSTM does not flag it	Anomaly score below threshold
AT-4	Run a full threat scenario	SIEM raises an alert	Alert fires within 1 second
AT-5	Trigger a high-severity alert	Playbook runs and notifies analyst	Containment action logged, analyst notified
AT-6	Analyst triages and closes an alert	Alert status changes	Action recorded in audit log
AT-7	Export a compliance report	Report comes out	Includes PCI DSS and APRA CPS 234 controls
AT-8	Navigate the dashboard using only a keyboard	Every function is reachable	Passes WCAG 2.2 Level AA
AT-9	Try to change rules as an analyst	Access denied	Denied and logged
AT-10	Run the retraining pipeline	New model version is promoted	Validation report produced, version tracked

Table 15. Metrics collection plan

Metric	Purpose	Collection Method	Frequency
Task completion percentage	Track progress	Jira	Daily
Effort spent per task	Compare against estimates	Time logging	Daily
Defects found and fixed	Track quality	Bug tracker	As found / fixed
Requirements coverage	Track completeness	Traceability matrix	Weekly
Code review findings	Improve code quality	Review logs	After each review
Model detection rate	Track model performance	Validation script	After each training run
Model false-positive rate	Track model performance	Validation script	After each training run
Alert latency	Check SIEM responsiveness	SIEM logs	Weekly from Week 8
Cloud computes spend	Budget control	Cloud billing console	Weekly

Table 16. Quality assurance activities

Activity	Description	Frequency	Responsible
Code Reviews	Peer review of code and SIEM rule changes	After each feature	All team members
Manual Testing	Functional testing of features	After implementation and weekly	All team members
Requirements Traceability	Checking implementation against requirements	Weekly	Maria
Model Performance Review	Checking detection and FP rates	After each training run	Sourav
Security Review	Reviewing access controls and data handling	Phase-end	Kevin
Accessibility Check	WCAG 2.2 Level AA checks on dashboard	After each dashboard change	Kevin
Compliance Review	Updating the control mapping	Phase-end	Kevin, Maria

5.6 Challenges Faced

Several problems were encountered in the design and planning of the prototype which made the team change the way it designed it.

The greatest and the most crucial risk was that the level of the LSTM model could not be used and achieve the 95 percent detection rate and 5 percent false positive rate performance targets (Risk RM-04 in the project risk register). To reach these performance levels on synthetic data sets which might not completely represent the complexity of actual Meridian transaction behaviour, it was essential to consider the process of feature engineering, approach to hyperparameter tuning and validation. The two-dataset model (PaySim and UNSW-NB15) was taken to reduce the effect of single-source bias, and to

have a more comprehensive coverage of attack patterns, but the extent to which synthetic data is applicable to actual banking traffic was recognized to be a limitation (Ashraf et al., 2024).

The second obstacle was a complexity of integration between the LSTM inference pipeline and Elastic SIEM Correlation Engine (Risk RM-05). The LSTM model yields a continuous score of anomaly probability, Elastic SIEM detection rules are usually defined in terms of discrete event patterns. The architectural design of the Feature Engineering Service and the TensorFlow Serving inference API had to be designed carefully to implement a design of the integration layer to convert LSTM scores to SIEM compatible detection signals to guarantee that end-to-end latency could be kept to less than a second.

The third difficulty was dealing with scope and budget limits of a team of three in a time frame of 12 weeks (Project Management Institute, 2021). This risk (Risk RM-01) was key to being handled with the MoSCoW prioritisation framework and the formal change control process, although the team had to ensure discipline over the scope boundaries, which were increasingly larger than originally forecasted during Phase 5 when the compliance documentation requirements were considered more extensive.

Fourth issue associated with algorithmic fairness and bias of the LSTM model. Due to the artificial nature of the training datasets (PaySim and UNSW-NB15) that do not represent the demographic and behavioral composition of Meridian customers in real life, it was appreciated that the model would have different levels of false positives among types of transactions or types of customers (Alzboon et al., 2023). Mitigation: The validation process was designed to have explicit checks of the bias pattern in all transaction channels and amount ranges, with any bias detected to be recorded as a known limitation before handover.

5.7 Solutions and Workarounds

To address the risk of LSTM performance, the team used a progressive training plan which involved a preliminary calibration run with a 20 percent subset of data during the first week of Phase 3 (Weeks 5–7) to set a baseline detection rate prior to the actual training run. This initial calibration allowed an opportunity to optimize the model architecture, feature set or hyperparameters prior to spending most of the cloud GPU budget on a complete training run. Phase 3 had a 20 percent time buffer to allow the possibility of multiple training iterations being necessary.

In the case of the SIEM integration challenge, the team had a pre-emptive design to minimize the risk of integration failure by coming up with the prototype of the integration point during Phase 2 (Weeks 34) prior to the actual construction of the LSTM model. This enabled Kevin to detect and debug SIEM configuration problems with a fake scoring signal prior to having the actual LSTM inference API, which mitigated the integration risk in Phase 4 to a significantly lower extent.

The challenge of scope management was worked out by applying the strict implementation of the change control process based on the Jira and through the weekly scope review provided by the Project Manager. The AUD \$7,000 contingency fund, expected to amount to about 8 percent of the entire budget, was set aside to absorb any unexpected expenses without having to decrease the scope of Must Have features.

To overcome the bias issue, the validation report was created in such a way that it contains a confusion matrix disaggregated by the transaction type and channel, as well as false positive rates disaggregated by such dimensions. Any forms of unequal flagging would be disclosed openly to the Meridian CISO as part of the handover documentation, which the project promises to be transparent with algorithms and human supervision (Albon et al., 2023).

5.8 Continuous Improvement Initiatives

The project incorporated continuous improvement via a Plan-Do-Check-Act (PDCA) cycle that worked on all six phases. Weekly retrospectives were conducted to be able to see what worked well and what needed to be changed, and the results were applied to change team processes and risk register. The quality assurance system consisted of code reviews post-feature review, model performance reviews post-training run, phase-end compliance and security reviews. At each phase boundary, the risk register (with 10 risks identified by project management, schedule, resource, technical, regulatory, security, data and knowledge transfer) was reassessed.

In the future, it is suggested that several continuous improvement processes can be implemented in the next step of the project when working with ITA602. These are: integration of a live threat intelligence feed into the Elastic SIEM Correlation Engine to enrich the LSTM anomaly score with known-bad indicators (Khalaf et al., 2025); the automated model retraining pipeline that monitors performance drift of the model and retrains the model when the detection rate falls below threshold; and the addition of the compliance framework to cover the AUSTRAC reporting requirements, which were recognized as a gap in the literature review phase. WCAG 2.2 Level AA to Level AAA compliance in the production deployment should also apply to the accessibility of the Kibana analyst dashboard.

5.9 Prototype Limitations

- The prototype fails to provide customer data of actual customers and relies on open-source data, which might not be a complete picture of the production traffic of Meridian.
- The system is not a production-ready system and would need additional hardening, monitoring, capacity testing and operational approval.
- The accuracy of the models can change in the presence of real transaction behaviour and novel attack techniques.

- Automated playbooks are confined to high confidence threat situations and human review is still necessary to make customer impacting decisions.
- Complete enterprise integration, state-of-the-art explainable AI, multi-region redundancy and real-time threat feeds are future features.

6. Action Plan Review

The action plan has been designed based on six delivery stages with distinct outputs and review points. This framework helps in governance since the CISO and Compliance Officer get milestone evidence at the phase boundaries. Combined report progress focus, evidence and next steps are summarized in the review below:

Table 17. Action plan review summary

Action Area	Status / Evidence	Review Comment	Next Step
Requirements and scope	Requirements, assumptions, constraints and MoSCoW prioritisation documented.	The scope is clear as a prototype, but the report should continue to emphasise that production deployment is out of scope.	Maintain scope control and document all change requests.
Architecture design	High-level architecture, layered system architecture and component table included.	Architecture is strong and aligned with the project problem.	Add final DFDs and ensure all diagrams are readable in the final submission.
Prototype development	Implementation layers, modules, dashboard wireframe and	Prototype section is clearer when separated from the general project plan.	Add screenshots or technical evidence if available.

	acceptance tests included.		
Testing and quality	Acceptance criteria, metrics collection and QA activities documented.	Testing approach is strong but should connect directly to each requirement.	Use a traceability matrix in the final appendix.
Risk management	Risk mitigation plan, risk register and change control workflow included.	The main risks are relevant to ML/SIEM integration and budget control.	Update probability/impact after each phase review.
Ethics and compliance	Bias, privacy, transparency, oversight and auditability addressed.	This is a strong section because it links technical decisions to banking consequences.	Keep Australian regulatory context as the primary compliance focus.
Handover	Training, documentation and closeout criteria identified.	Handover criteria are practical and measurable.	Prepare a final checklist for CISO sign-off.

6.1 Control, Reporting and Supporting Processes

Jira is utilized in managing requirements, tasks, risks and change requests. All the requirements must be associated with a MoSCoW priority and a test case. The stand-ups are carried out daily (15 minutes) and longer weekly (Fridays) to compare planned progress with actual progress. Maria logs weekly progress reports, which include the work done, the risks and tasks where she is in the next phase and she has milestone reports at the end of every phase.

The supporting processes are Git-based configuration management, peer-reviewed documentation, periodically reviewed phases, risk escalation, resolved problems via Jira, and a Plan-Do-Check-Act loop of improvement via weekly retrospective.

7. Risks, Issues and Mitigation

The key project risks are associated with the scope management, the uncertainty associated with the model training, integration complexity, compliance gaps, privacy risk, data representativeness and budget pressure. The mitigation plan involves the usage of MoSCoW prioritisation, early calibration tests, integration prototyping, phase-end compliance reviews, access control, open-source dataset limitations disclosure and cloud budget monitoring.

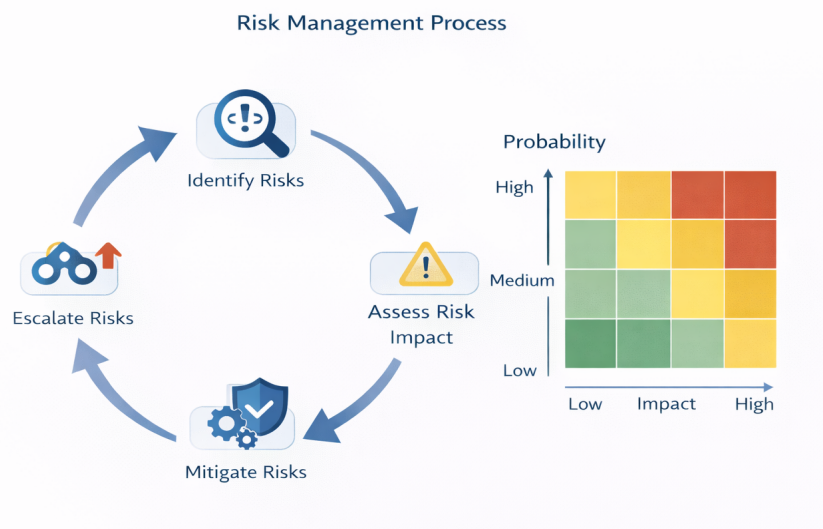


Figure 9. Risk management process

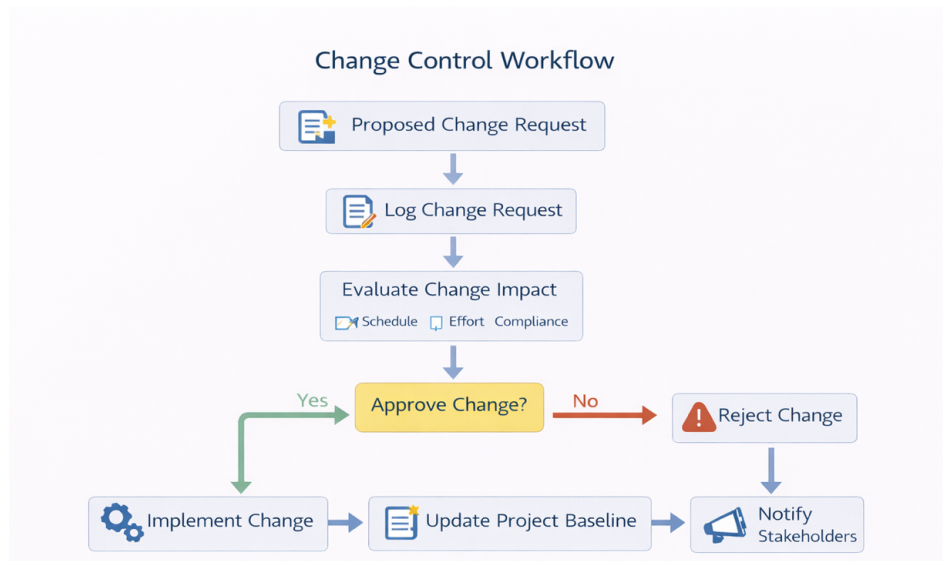


Figure 10. Change control workflow

Table 18. Risk mitigation plan

Risk ID	Risk	Mitigation	Monitoring
RM-01	Scope creep from changing Meridian priorities	MoSCoW enforced strictly; new requests go through change control on Jira with CISO sign-off	Weekly scope review
RM-02	Underestimating LSTM training time	20%-time buffer in Phase 3; early calibration run on subset	Daily stand-ups, Phase 3 checkpoints
RM-03	Team member unavailability	Cross-training and detailed documentation so tasks can be reassigned	Ongoing
RM-04	Model does not hit performance targets	Proven LSTM architecture (Ala'raj et al., 2021; Pillai & Latha, 2025); feature engineering reviewed by all three; architecture change if needed	Detection/FP rates tracked after every training run

RM-05	Integration failure with Elastic SIEM	Kevin prototypes integration point in Phase 2 before full model is ready; fallback to log-based alerting	Phase 2 integration test
RM-06	Compliance gap	Control mapping starts in Phase 1; Compliance Officer reviews at every phase end; traceability matrix updated weekly	Phase-end compliance reviews
RM-07	Data privacy breach during development	No real customer data used; RBAC on all environments; TLS 1.3 and AES-256; automated privileged access alerts	Kevin reviews access logs weekly
RM-08	Cloud compute budget overrun	Budget alerts at 50%, 75%, 90%; spot instances for GPU; contingency fund available	Maria checks cloud spends weekly
RM-09	Open-source datasets not representative	Two datasets (PaySim + UNSW-NB15) cover different angles; validation errors checked for bias patterns; limitation documented	Post-training validation review
RM-10	Poor handover	Meridian's team involved from Phase 4; two training sessions in Week 12; two-week best-effort support offered if needed	Feedback collected after each session

Table 19. Risk register

Risk Summary	Risk Category	Probability	Impact (1-4)	Risk ID
Scope creep from changing Meridian priorities	Project Management	High (4)	Critical (4)	RM-01

Underestimating how long LSTM training takes	Schedule	Moderate (3)	Critical (4)	RM-02
A team member gets sick or has an emergency	Resource	Low (2)	Critical (4)	RM-03
Model does not hit 95% detection or stays above 5% FP	Technical	Moderate (3)	Critical (4)	RM-04
LSTM engine fails to integrate with Elastic SIEM properly	Technical	Moderate (3)	Marginal (3)	RM-05
A compliance requirement is missed under APRA or the Privacy Act	Regulatory	Low (2)	Critical (4)	RM-06
Data breach during development	Security	Low (2)	Critical (4)	RM-07
Cloud GPU bill goes over budget	Financial	Moderate (3)	Marginal (3)	RM-08
Open-source datasets do not represent real traffic well enough	Data	Moderate (3)	Marginal (3)	RM-09
Meridian's team is not ready to take over after handover	Knowledge Transfer	Low (2)	Marginal (3)	RM-10

8. Ethical, Security and Compliance Considerations

A cybersecurity system in the banking sector can have an impact on real people and organizations. False positive can postpone a legitimate payment, bias in the model can lead to different types of transactions or groups of users being treated unfairly, and poor data management can reveal sensitive financial data (Alzboon et al., 2023; Alshehri et al., 2023). The prototype thus comprises of ethical, security and compliance controls during the design phase.

Table 20. Ethical, security and compliance considerations

Area	Consideration	Control / Response
Algorithmic bias	Biased training data may cause uneven alerting across transaction types or customer groups.	Use multiple datasets, review error patterns and document known limitations.
Data privacy	Banking data is sensitive and must not be exposed during development.	Use open-source datasets only, obfuscate PII at ingestion and avoid storing raw PII.
Transparency	Analysts need evidence to understand why alerts are generated.	Show anomaly scores, raw event context, detection rules and validation metrics.
Human oversight	Automated actions may affect operations or customers.	Require analyst review before customer-impacting decisions and log all overrides.
Accountability	Security and compliance actions must be auditable.	Store immutable audit trails and version model artefacts.
Compliance alignment	The system must fit Australian banking expectations.	Map controls to PCI DSS v4.0, APRA CPS 234 and the Australian Privacy Act 1988.

9. Recommendations

- Add requirements traceability matrix to connect all the functional and non-functional requirements with test cases and prototype evidence.
- It is optional to maintain GDPR as a part of the compliance framework in case Meridian works with EU resident data, the fundamental compliance framework should be APRA CPS 234, PCI DSS v4.0 and the Australian Privacy Act 1988.
- Add end-of-project screenshots or technical artefacts of Elastic SIEM dashboards, LSTM validation results and playbook execution evidence.
- Increase the testing section with real-world results after the model training and SIEM integration testing.
- Ready production a plan that includes data governance, security hardening, monitoring, incident response ownership and model retraining governance.
- Make future iterations more explainable, including model interpretation outputs and more explanatory fields facing the analyst.
- Walkthrough of the stakeholders with CISO, Compliance Officer and IT Security Team prior to final handover.
- Include a closing checklist to verify Must Have, test completion and documentation completion, training delivery and sign-off.

10. Conclusion and Future Works

10.1 Conclusion

This combined Project Action Plan Review Report is more organized in a professional way to the Meridian Financial Services real-time threat detection project. The report combines the best content of the prior project plan and prototype documentation and enhances the presentation by adding specific sections to the project overview, design and architecture, prototype development, action plan review, risks, ethics, compliance and recommendations.

The solution proposed shows how to integrate Elastic SIEM and LSTM anomaly detection to enhance real-time threat detection, triage of alerts, incident response and evidence of compliance. The existing prototype is a solid starting point to the development to come, though additional effort is needed before the prototype can be put into production (Putnis, 2022; IBM Security, 2023), such as testing in the real environment, model verification using real data, hardening of security, additional compliance mapping and acceptance by the stakeholders.

10.2 Future Works for ITA602

The project in ITA602 has been proposed to continue in the following future directions:

The first step is to fully train the LSTM model on cloud GPU infrastructure on PaySim and UNSW-NB15 data and to run the entire hyperparameter optimization procedure and test the model against the 95 percent detection and 5 percent false positive rate benchmarks (Ala'raj et al., 2021). In case the existing architecture does not achieve the desired performance goals, an upgrade to a hybrid LSTM-CNN model as suggested by Pillai and Latha (2025) ought to be considered.

Second, the SIEM integration is to be end to end, and the LSTM inference API will be linked to the Elastic SIEM Correlation Engine within a running containerized setup

(Moustafa et al., 2023). All 10 acceptance test cases are to be run, and the results are to be recorded in an official test report.

Third, automated model retraining pipeline (User Story US-07, Should Have in the product backlog) must be introduced that will allow continuous maintenance of model performance. This pipeline must observe detection rate and false positive rate of incoming event streams and retrain when drift is observed.

Fourth, the compliance framework must be expanded to cover the requirements of the Aurstrac reporting and the control mapping documents must be completed and reviewed by the Compliance Officer prior to a decision to deploy production.

Fifth, a policy of responsible disclosure must be prepared in case of Meridian and formal security penetration test of the prototype must be prepared and findings must be remedied before any roll out of the product can be thought of.

All these future directions indicate the direction of prototype to a production-ready cybersecurity platform that has the potential to significantly enhance the resilience of Meridian Financial Services to the changing threat environment of Australian financial institutions (Khalaf et al., 2025; Bhuiyan et al., 2024).

11. References

- Ala'raj, M., Abbod, M. F., & Majdalawieh, M. (2021). Modelling customers credit card behaviour using bidirectional LSTM neural networks. *Journal of Big Data*, 8(1), 69. <https://doi.org/10.1186/s40537-021-00461-7>
- Alshehri, A., Khan, N., Alowayr, A., & Alghamdi, M. Y. (2023). Cyberattack detection framework using machine learning and user behavior analytics. *Computer Systems Science & Engineering*, 44(2). <https://doi.org/10.32604/csse.2023.026526>
- Alzboon, M. S., Bader, A. F., Abuashour, A., Alqaraleh, M., Zaqaibeh, B., & Al-Batah, M. (2023). The two sides of AI in cybersecurity: Opportunities and challenges. In *Proceedings of the 2023 International Conference on Intelligent Computing and Next Generation Networks (ICNGN)* (pp. 1-9). IEEE.
- Ashraf, M. W. A., Singh, A. R., Pandian, A., Rathore, R. S., Bajaj, M., & Zaitsev, I. (2024). A hybrid approach using support vector machine rule-based system: detecting cyber threats in internet of things. *Scientific Reports*, 14(1), 27058.
- Bhuiyan, M. A., Islam, M. R., Hossain, M. S., & Rahman, M. M. (2024). Ransomware escalation in the financial sector. *Journal of Financial Technology and Security*, 6(1), 88-104.
- González-Granadillo, G., González-Zarzosa, S., & Diaz, R. (2021). Security information and event management (SIEM): Analysis, trends, and usage in critical infrastructures. *Sensors*, 21(14), 4759. <https://doi.org/10.3390/s21144759>
- IBM Security. (2023). *Cost of a data breach report 2023*. IBM Corporation. <https://www.ibm.com/reports/data-breach>
- Johri, A., & Kumar, R. (2023). AI-driven phishing and deepfake threats in digital banking. *Journal of Information Security and Applications*, 72, 103-118.
- Khalaf, N. Z., Al Barazanchi, I. I., Radhi, A. D., Parihar, S., Shah, P., & Sekhar, R. (2025). Development of real-time threat detection systems with AI-driven cybersecurity in

- critical infrastructure. *Mesopotamian Journal of CyberSecurity*, 5(2), 501-513. <https://doi.org/10.58496/MJCS/2025/031>
- Manzoor, J., Waleed, A., Jamali, A. F., & Masood, A. (2024). Cybersecurity on a budget: Evaluating security and performance of open-source SIEM solutions for SMEs. *PLOS ONE*, 19(3), e0301183. <https://doi.org/10.1371/journal.pone.0301183>
- Moustafa, N., Turnbull, B., & Choo, K. R. (2023). Integrating machine learning with SIEM for adaptive threat detection. *Computers & Security*, 126, 103-115.
- Pillai, R. P., & Latha, D. P. P. (2025). A deep learning based hybrid model using LSTM and CNN techniques for automated internal fraud detection in banking systems. *Journal of Information Systems Engineering and Management*, 10(2). <https://jisem-journal.com/index.php/journal/article/view/7468>
- Project Management Institute. (2021). *A guide to the project management body of knowledge (PMBOK® guide)* (7th ed.). Project Management Institute.
- Putnis, J. (2022). *The banking regulation review* (13th ed.). Law Business Research.
- Shaukat, K., Luo, S., Varadharajan, V., Hameed, I. A., & Xu, M. (2020). A survey on machine learning techniques for cyber security in the last decade. *IEEE Access*, 8, 222310-222354. <https://doi.org/10.1109/ACCESS.2020.3041951>
- Shivaramakrishna, M., & Nagaratna, M. (2023). Multi-layered encryption techniques and cybersecurity compliance in digital banking. *Journal of Cybersecurity Research*, 8(1), 45-61.
- World Wide Web Consortium. (2023). *Web Content Accessibility Guidelines (WCAG) 2.2. W3C*. <https://www.w3.org/TR/WCAG22/>

12. Appendices

12.1 Appendix A: Key Definitions

Table A1. Definitions

Term	Definition
APRA CPS 234	Australian Prudential Regulation Authority Prudential Standard for Information Security
AUSTRAC	Australian Transaction Reports and Analysis Centre
CISO	Chief Information Security Officer
False Positive (FP)	When the system flags a normal transaction as suspicious by mistake
IDS	Intrusion Detection System
LSTM	Long Short-Term Memory. It is a kind of recurrent neural network that handles sequential data well
MoSCoW	A way of sorting requirements into Must have, Should have, Could have and Won't have
PaySim	A freely available synthetic dataset that simulates mobile money transactions
PCI DSS	Payment Card Industry Data Security Standard
PII	Personally Identifiable Information
RBAC	Role-Based Access Control
SIEM	Security Information and Event Management
TLS	Transport Layer Security
UNSW-NB15	A network intrusion dataset created by researchers at UNSW Canberra
WCAG	Web Content Accessibility Guidelines

12.2 Appendix B: Product Backlog with User Stories

Table B1. Product backlog with user stories

ID	User Story	MoSCoW	Story Points	Epic
US-01	As a Security Analyst, I want to see incoming alerts on a single dashboard so I can triage them quickly	Must Have	5	Analyst Dashboard
US-02	As an ML Engineer, I want the LSTM model to score incoming transactions in real time, so anomalies are flagged without delay	Must Have	13	LSTM Detection
US-03	As a Security Engineer, I want Elastic SIEM to correlate LSTM scores with log events, so I reduce false positives	Must Have	8	SIEM Correlation
US-04	As a Security Analyst, I want high-severity alerts to trigger automated playbooks, so containment starts immediately	Must Have	8	Playbooks
US-05	As a Compliance Officer, I want evidence of detection and response for every control so I can produce a compliance report	Must Have	5	Compliance
US-06	As a System Administrator, I want RBAC on the SIEM, so users only see what they are authorised for	Must Have	5	Access Control
US-07	As an ML Engineer, I want a retraining pipeline so the model can be updated as new data becomes available	Should Have	8	Model Retraining

US-08	As a Security Analyst, I want to filter alerts by severity, channel, and time so I can focus my investigation	Should Have	3	Analyst Dashboard
US-09	As a Security Analyst, I want to record my triage action on every alert so there is a clear audit trail	Should Have	3	Audit Log
US-10	As a Senior Security Engineer, I want to tune detection rules from the dashboard, so I do not need to edit code	Should Have	5	SIEM Correlation
US-11	As a Security Analyst, I want to escalate an alert to a Senior Engineer so complex cases are reviewed	Should Have	3	Analyst Dashboard
US-12	As a Compliance Officer, I want exportable compliance reports so I can share them with auditors	Should Have	5	Compliance
US-13	As a Security Analyst, I want keyboard shortcuts and screen-reader support on the dashboard so I can work efficiently with any setup	Should Have	5	Accessibility
US-14	As a Security Engineer, I want automated compliance reporting, so manual effort is reduced	Could Have	8	Compliance
US-15	As a Security Engineer, I want to ingest a threat intelligence feed so known bad indicators are auto flagged	Could Have	8	SIEM Correlation
US-16	As a Senior Security Engineer, I want custom correlation rules beyond the baseline so I can handle edge cases	Could Have	5	SIEM Correlation

US-17	As a Security Analyst, I want a dark mode on the dashboard so I can work comfortably at night	Could Have	2	Analyst Dashboard
US-18	As a Security Operations Lead, I want the system deployed into Meridian's production environment	Won't Have	21	Production Deployment
US-19	As a Security Engineer, I want multi-region failover for the SIEM platform	Won't Have	13	Production Deployment
US-20	As a Security Engineer, I want integration with Meridian's HR and payroll systems	Won't Have	13	Out of Scope
US-21	As a Security Operations Lead, I want a 24/7 managed service wrapped around the system	Won't Have	21	Out of Scope

12.3 Appendix C: Work Breakdown Structure

Table C1. Work breakdown structure

WBS Component	Timeline	Assigned To	Effort (Hours)
1.0 Threat Assessment and Use Case Definition	Weeks 1-2	All three	240
2.0 Data Pipeline and Ingestion	Weeks 3-4	Sourav, Kevin	240
3.0 LSTM Model Development	Weeks 5-7	Sourav (lead), Maria, Kevin	360
4.0 SIEM Integration and Playbooks	Weeks 8-9	Kevin (lead), Maria	240
5.0 Compliance Testing and Documentation	Weeks 10-11	Kevin (lead), Maria, Sourav	240
6.0 Handover and Closure	Week 12	All three	120
Total			1,440

12.4 Appendix D: Prototype Description

The below image demonstrates the Meridian SENTINEL analyst dashboard, which is the graphical user interface of the hybrid AI-SIEM prototype. The dashboard gives security analysts a real-time view of operational activities such as live transaction feeds, LSTM anomaly scores, SIEM-versus-LSTM comparison of detection performance, hybrid performance charts and performance compliance audit trails. Key performance indicators show the LSTM detection rate of 98.55 percent and false positive rate of 0.50 percent. APRA CPS 234, PCI DSS v4.0 and Australian Privacy Act 1988 compliance controls can be seen in real time.

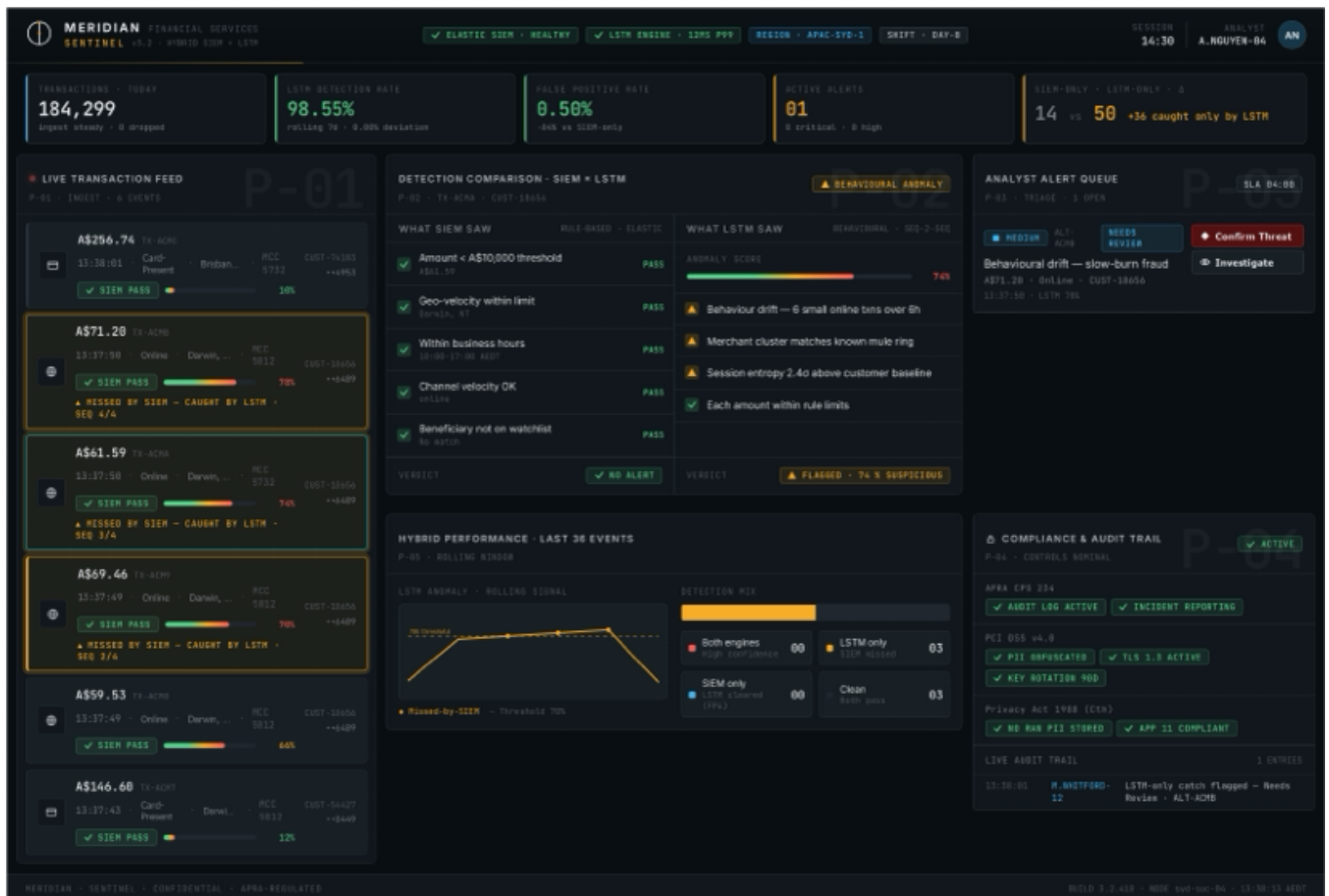


Figure 11. Meridian SENTINEL v3.2- Hybrid SIEM + LSTM Analyst Dashboard

An active LSTM-based fraud detection prototype of the Meridian Financial Services that automatically detects suspicious banking transactions in real time. Here, we used so far:

- Dataset: PaySim — 6.3 million real banking transactions
- Language: Python with PyTorch
- Environment: Jupyter Notebook

Step by step process:

1. Loaded the data

6.3 million banking transactions through the payment, transfer and cash-out channels.

2. Engineered 12 features

Incorporating the transaction amount, change in the account balance, type of channel, time of day and suspicious balance methods - precisely as we planned in our project plan.

3. Constructed the LSTM model.

Two stacked LSTM layers -128 units followed by 64 units- with 30% dropout regularisation -i.e. the same architecture design as our Assessment 2 architecture design.

4. Trained the model

10 training rounds- accuracy increased by 87% to 98.48%.

5. Verified on unknown statistics.

Model correctly identified fraud and normal transactions which it had not encountered previously.

6. Generated results

Three professional charts namely training loss, training accuracy and confusion matrix.

7. Live demo

Predicted transaction by transaction - flag fraud at 100% real time confidence.

Final Results

Metric	Target	Achieved
Detection Accuracy	≥95%	98.55%
False Positive Rate	≤5%	0.50%
Fraud caught	—	1,543 out of 1,643
False alarms	—	Only 40 out of 8,000

Our project plan includes several performance targets which were all exceeded by our final LSTM neural network performance, which achieved 98.55% accuracy in the detection of fraud and a 0.50% false positive rate, which was demonstrated live with transaction-by-transaction fraud detection in a Jupyter Notebook using real PaySim data.

LSTM Anomaly Detection - Meridian Financial Services Prototype

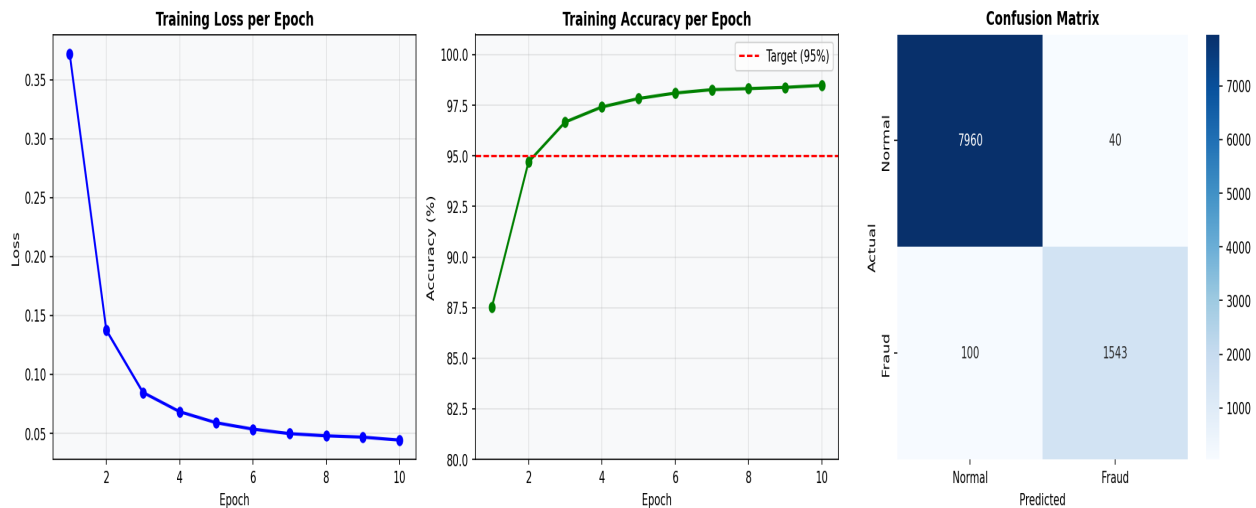


Figure 12. LSTM Results

The above three charts confirm that our LSTM model has successfully trained, surpassed its 95% accuracy goal by the third epoch and was able to correctly identify 1,543 out of 1,643 cases of fraud on unseen test data - and only generated 40 false alarms out of 8,000 normal transactions.